

FFAR Final Progress Report

Grant Information	
Grant ID	EggTech-0000000039
Award Program	EggTech Challenge
Project Title	“Rapid and Non-invasive Egg Sex Identification Using Artificially Intelligent Platform”
Reporting Period	10/10/2019 - 12/31/2022
Period Budget	
Project Director/Principal Investigator Information	
Full Name	Abdenmour Abbas
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Grantee Organization Information	
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Authorized Signing Official Information	
Name	Kevin McKoskey, April Coon
ASO Title	Grand manager
Email	awards@umn.edu (office), john5885@umn.edu (April Coon)
Phone	612-624-5599 (office), 612-624-7021 (April Coon)

1. General Information

1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the U.S., please list the city and country.

The work was conducted at:
2004 Folwell ave, 55108 Saint Paul, MN
Congressional District: MN-004

1.2. How many new jobs were created by the grant during this reporting period?

Four jobs were created with this grant:

- A postdoctoral research assistant in mass spectrometry (Dr. Quan Dong)
- A postdoctoral research assistant in machine learning (Dr. Kaige Zhang)

- **One graduate research assistant (Mr. Charles Broadbent)**
- **One undergraduate research assistant (Ms. Maddi King)**
- **One Researcher 5 (Akli Zarouri)**

1.3. How many jobs were maintained by the grant during this reporting period?

Two jobs:

- **One Researcher 5 (Akli Zarouri)**

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

2. Accomplishments

2.1. What were the goals/specific aims of the project for this reporting period? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 500-word limit)

The goal of this project is to demonstrate the proof-of-concept

Objective 1: Data collection of volatile organic compounds (VOCs) and geometrical characteristics from thousands of eggs during the first days of incubation.

Completion: 100%

Objective: 2: Identification of VOCs signatures/VOC biomarkers and egg geometries specific to male and females eggs.

Completion: 100%

Objective 3: Applying machine-learning algorithms to analyze the collected data to identify VOCs and geometrical patterns specific to male-containing eggs.

Completion: 100%

2.2. Have any of the major goals/specific aims or milestones for this reporting period changed since the award or previous report? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 300-word limit)

NO

2.3. What was accomplished under the goals/specific aims or milestones for this reporting period? For this reporting period describe: 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include a discussion of stated goals not met. As the project progresses, the emphasis in reporting in this section should shift from reporting activities to reporting accomplishments. In the response, emphasize the significance of the findings to the scientific field. Include approaches taken to ensure robust and unbiased results. (up to 1,500-word limit)

Experimental work:

The experimental work is performed in three major tasks

- 1- Egg Collection: Hy-line LLC. agreed to donate 180 white or brown eggs biweekly. The eggs are picked up from the hatchery site of the Hy-line company in Jackson, MN. The egg tray is properly labeled and covered to prevent from introducing interferences of trace chemicals in volatile organic compounds data collection. Once the fertilized chicken eggs are received, we put them into one of the OVA Easy 190-Advanced Cabinet for the short period of storage. However, this storage cannot exceed 3-4 days with requirements of 18-20 °C of 75% humidity. On the day when the fertilized eggs arrive, and before data collection, all the received eggs have to go through the fertility checking by using Egg Candler. The fertilized eggs will be selected for the experimental groups and the non-fertilized eggs will be used for the control group data recording. The eggs will then undergo incubation for 7 days. (Figure 1).

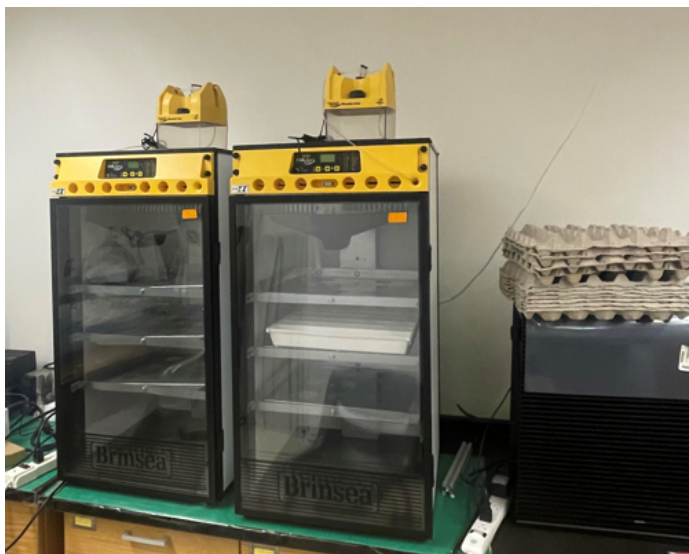


Figure 1. chicken egg incubators used in this project.

- 2- Egg analysis: The eggs are then organized for data collection every day during the first 7 days of incubation. Data collection include 6 features:
 - Morphological features (egg geometry)
 - Physical (Weight)
 - Metabolic features (volatile organic compounds)
- 3- Egg gender confirmation: The gender of the chicken is confirmed by running the DNA nucleic acid extraction, PCR, and gel electrophoresis. We have developed a protocol that

Results and Achievements:

While the weight and geometry parameters have yielded no conclusive results, the work resulted promising results for the VOC fingerprints and led to the in the development of several protocols and other findings, including:

- 1- New protocols: We developed a protocol for the extraction of nucleic acids from the chicken brain.

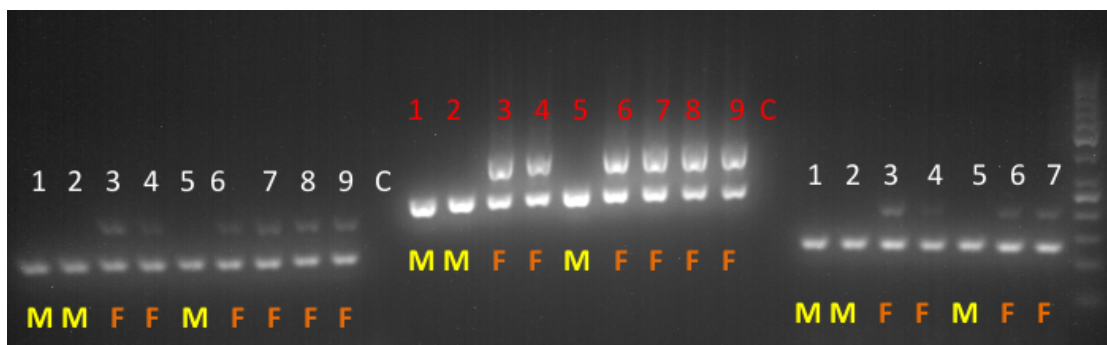


Figure 2. Gel electrophoresis of XhoI and 18S genes from chicken brain extract, used for egg gender confirmation.

- 2- Detection timing: We identified that the accuracy of detection reaches 97-100% when measurements are conducted on day2, 4 or 6. This is explained by the fact that Day 2 correspond to heart formation, Day 2 to respiration and Day 3 to Vitelline growth, all of which enhances the metabolic activity and thus VOC formation.
- 3- We have developed a protocol to confirm egg gender identification, using DNA nucleic acid extraction from the chicken brain, followed by PCR and gel electrophoresis. The method relies on genetic identification of XhoI (415 bp) and 18S (256bp) genes. Female eggs have both genes, while males have only 18S gene (**Figure 2**).
- 4- **VOC biomarkers:** We successfully identified 5 VOC biomarkers for egg gender: $C_{11}H_{17}O+$, C_9H_9+ , $C_{12}H_{15}O+$, $C_{14}H_{15}+$, $C_6H_{13}+$ (**Figure 3**). The acquisition of a gas chromatography instruments and its use in combination with the PTR-MS enabled the identification of the VOC as: tert-Butyl(methyl)phenyloxidanium, Cyclopropylcyclohexatriene, 1-(4-Methoxyphenyl)-4-pentenyl, Cyclohexadienylum, and Hexane-1-ylum.

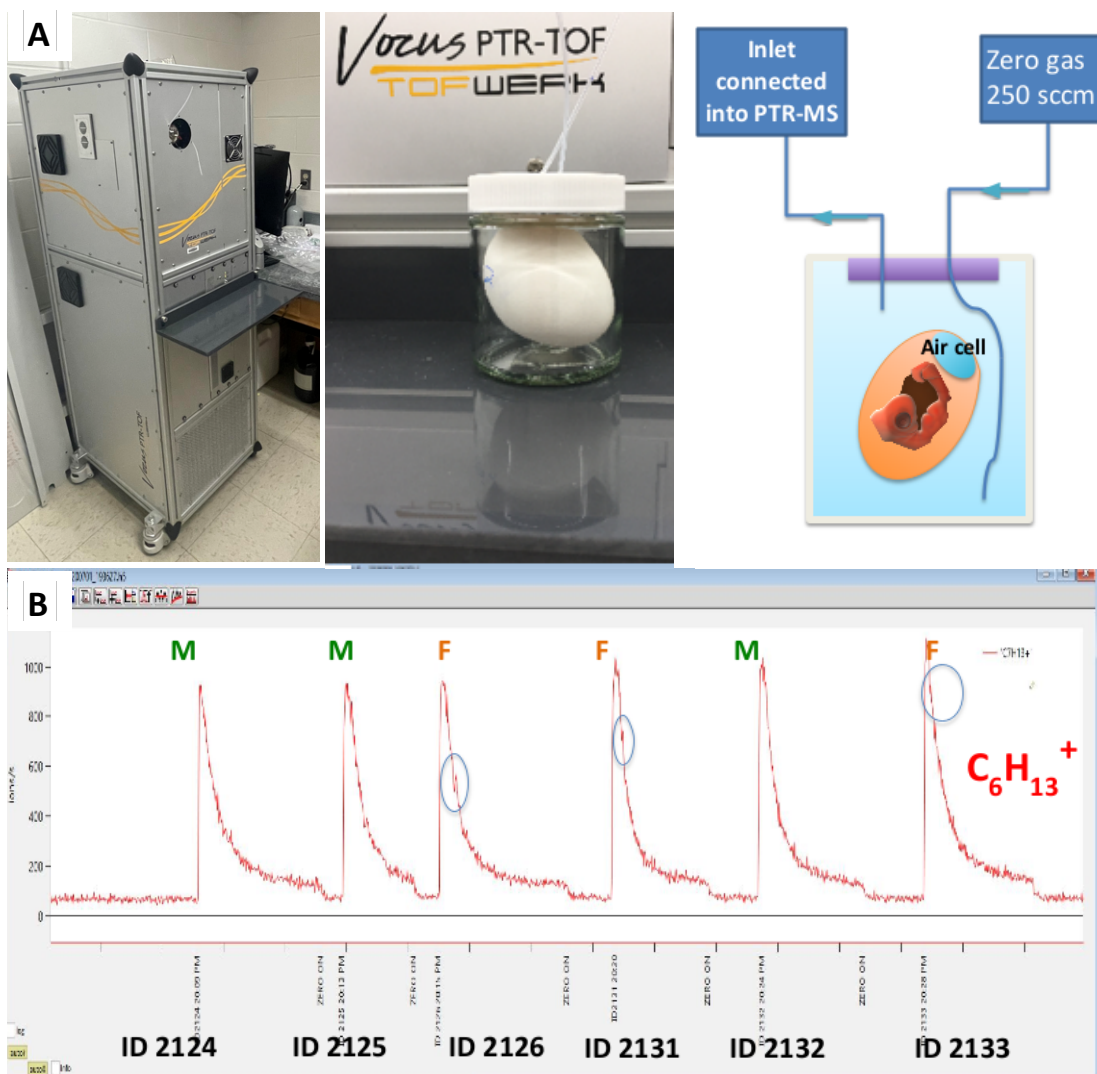


Figure 3. Egg VOC collection and analysis using PTR-MS during incubation. (A) VOC collection using PTR-MS. (B) PTR-MS spectra of $C_6H_{13}^+$ as potential biomarker for egg gender identification.

2.3.1. Aside from the accomplishments outlined above, have there been any other significant impacts resulting from the work under this grant? Please describe any broader impacts such as:

- *Tools developed*
- *Benefits to policy*
- *Benefits to future research*
- *Benefits to food, the food system or agriculture*
- *Broader economic or health benefits*

If there are no broader impacts to date, state “Nothing to Report.”

Nothing to report

2.4. *Describe challenges or delays encountered during the reporting period and actions or plans to resolve them. Only describe significant challenges that may impede the research and emphasize their resolution. (up to 500-word limit)*

There were two major challenges that delayed the project:

- a- **Hires:** Two of the postdocs working on the project (Dr. Zhang and Dr Dong) went back to China, and hiring replacement during the pandemic has been difficult, especially high skilled people in machine learning or mass spectrometry. International applicants were hesitant to travel to the US during the pandemic, but the conditions are improving and there seems to be more willingness to travel currently.
- b- **Equipment:** This project relied mainly on the PTR-MS equipment that allows quantification of different VOCs/gases released by the eggs. However, after identifying VOC fingerprints as potential biomarkers for egg gender, we couldn’t identify the chemical structure VOCs due to lack of a gas chromatography. The updated budget request includes the purchase of a Fast GC that can be integrated directly with the PTR-MSA equipment and enable both quantification and identification.

2.5. *Have there been any changes in scientific approach or reasons for change? If so, what are the changes? Remember, changes to the approved scientific approach must be pre-approved by FFAR. (up to 500-word limit)*

The scientific approach remains the same.

2.6. *What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)*

- **A postdoctoral research assistant in mass spectrometry (Dr. Quan Dong)**
- **A postdoctoral research assistant in machine learning (Dr. Kaige Zhang)**
- **Two graduate research assistants (Mr. Charles Broadbent, and Vinni Novi)**
- **One undergraduate research assistant (Ms. Maddi King)**

2.7. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 300-word limit)

- A postdoctoral research assistant in mass spectrometry (Dr. Quan Dong)
- A postdoctoral research assistant in machine learning (Dr. Kaige Zhang)
- Two graduate research assistants (Mr. Charles Broadbent, and Vinni Novi)
- One undergraduate research assistant (Ms. Maddi King)

2.8. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 500-word limit)

- **News Media:** The work has been mentioned in the Wall Street Journal article: “A Technology Race to Stop the Mass Killing of Baby Chicks”, May 29, 2021. <https://www.wsj.com/articles/a-technology-race-to-stop-the-mass-killing-of-baby-chicks-11622300400>
- Established a connection with a turkey producing company in Willmar, MN “Life Sciences Innovations” (contact: Bob Poe <Bob.Poe@life-scienceinnovations.com>). The company expressed interest in conducting a similar sexing study on turkey. We have already met 3 times and we are planning to conduct a study on their production site.
- We have been contacted by UK-based NGO, Verify Humanity. They are interested in collaborating for an EST (Egg Sex Tech) Campaign. Once we demonstrate the technology in the field, we plan to work with Verify Humanity to raise awareness about technology availability.

2.9. What do you plan to do during the next reporting period to accomplish the goals of the approved project? Briefly describe what you plan to do during the next reporting period to accomplish the project goals and objectives. Discuss efforts to ensure the approach is scientifically rigorous and results are robust and unbiased. Remember that significant changes to the approved goals and objectives and project scope require prior FFAR approval. Include any important modification to the original goals and provide justification for the change. (up to 500-word limit)

The biggest achievement of this project is the identification of 5 VOC markers for egg gender, namely tert-Butyl(methyl)phenyloxidanium, Cyclopropylcyclohexatriene, 1-(4-Methoxyphenyl)-4-pentenyl, Cyclohexadienylum, and Hexane-1-ylum.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models software, curricula, instruments or equipment, intervention, etc.) produced during this reporting period resulting directly from the FFAR award.

This reporting period resulted in the production of:

- *A new protocol for egg gender genetic identification using PCR and gel electrophoresis*
- *A new protocol for the extraction of nucleic acids from chicken brains.*
Publications and conferences are not planned until the full demonstration of the technology.

3.2. Please provide a list of citations for the information products produced during this period.

See question 3.1

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the reporting period resulting directly from the FFAR award? If yes, please provide citation.

See question 3.1

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

See question 3.1

3.5. Have inventions, patent applications, and/or licenses resulted from the award during this reporting period? If yes, indicate the invention, patent application(s), and/or license(s).

Not yet, but it's expected.

3.6. Are any of the information products produced during this reporting period confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if

*(and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely.
(up to 500-word limit)*

Yes, the list of VOC marker is confidential information.

3.7. *Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?*

4. Data Management

4.1. *During this period, did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated in this reporting period.*

Yes, the project generated significant data, including:

- ***PTR-MS spectra of egg VOC during incubation***
- ***Gel electrophoresis images***
- ***Egg weight distribution***
- ***Egg geometry measurements***
- ***Machine learning algorithms***

4.2. *If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.*

All the data sets are related to egg characterization during incubation for 7 days.

4.3. *Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Upload copies of records and a simple file inventory, if necessary, in a compressed folder.*

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is asignificant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I willnot claim otherwise.

PI Full Name: Abdenmour Abbas

PI Signature: -



Date: 03/02/2023

Authorized

Signing Official (ASO): _____

ASO Signature: _____ Date: _____